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Unit 3 : Session 9 : SLO 1 | Describe MVC Architecture in GUI Development

I. Fill in the Blanks

1. View – the View is responsible for presenting the user interface.
2. Controller – it listens to user actions and modifies the Model.
3. Controller – an ActionListener attached to a button plays the Controller role in Swing MVC.
4. reusability – MVC promotes code reusability (separation of concerns) and testability.

II. True or False

1. FALSE – the Controller updates the Model; the View is refreshed from the Model, not updated directly by the Controller bypassing it.
2. TRUE – MVC separates data (Model), presentation (View) and input handling (Controller), which is the separation of concerns.

III. Multiple Choice

1. C – the Model contains and manages the application data and business logic.
2. C – the MVC control flow is View → Controller → Model → View.

IV. Programming

1. Identify the output: A – each button click increments the counter in the Model and the label is updated with the new value.
2. Identify the missing part: C – `view.updateDisplay(model.getData());` after the Model changes, the View must be refreshed from the Model to complete the MVC cycle.